

CHRONIC HEPATITIS B

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INTRODUCTION

Hepatitis means inflammation of the liver. Hepatitis B essentially means an infection of the liver with the Hepatitis B Virus (HBV). The HBV is transmitted by blood or body fluids of an infected person to another who does not have immunity against it. A HBV infection can be cleared by the body especially in a healthy adult (acute Hepatitis B) but in some cases the infection progress to become a long term infection (chronic Hepatitis B) as the body is not able to clear it.

Most patients who contracted acute Hepatitis B might not develop any symptoms or only develop non-specific symptoms. Some can become very ill, but the likelihood of liver failure is less than 1%. On the other hand, patients with chronic Hepatitis B infection (also known as Hepatitis B carriers) are usually well looking and can unknowingly transmit the virus to others.

In chronic Hepatitis B, the virus might frequently flare up. The liver is susceptible to damage by the chronic Hepatitis B and the extent of damage is dependent on the liver's ability to repair itself and the body immune system to control the infection. The complication of the damage includes liver scarring (medically known as liver fibrosis), liver failure and liver cancer.

Liver cancer is known as a silent killer because the majority of patients do not have symptoms in the early stages. In the late stages, symptoms occur when the tumour is large and start to impair the function of the liver or cause space occupying symptoms such as pain and abdomen swelling.

There are drug treatments for Hepatitis B but no cure. Fortunately, there is a vaccine. According to the World Health Organization, the Hepatitis B vaccine is 95% effective in preventing children and adults from developing chronic Hepatitis B infection if they have not yet been infected. The vaccine is easily administered in a series of 3 intramuscular doses for adults. Hepatitis B immunisation can hence help to prevent the prevalence of primary liver cancer.

Epidemiology

Up to 2 billion people have been infected with Hepatitis B world-wide and 300 million are chronically infected. Countries in Asia, the Pacific Islands and Africa have moderate to high prevalence of chronic Hepatitis B.

In Singapore about 4% (1 in 25 persons) of the population are chronic Hepatitis B carriers which poses a significant cost to Singapore society as a whole. Hepatitis B is estimated to cause 60 to 80 per cent of primary liver cancers worldwide based on statistics from the World Health Organization. In Singapore 1,855 primary liver cancers were diagnosed between 1988 and 2002. Key findings from the National Sero-prevalence Survey 2005 conducted by the Ministry of Health showed that 59.3% of the population aged 30 to 74 years and 57.2% aged 30 to 44 years were not immune to the HBV.

Natural History of Chronic Hepatitis B

The likelihood of developing chronic hepatitis B in individuals is higher in those infected perinatally (90%) compared to those infected in adulthood (1%). This is due to differences in the maturity and competence of the immune systems during the respective ages. Most infections acquired in Singapore are perinatal or during early childhood, though the incidence is much lower due to the advent of the hepatitis B vaccination programme in 1987.

The natural history of perinatal and childhood-acquired infection is generally described to have three phases.

The first phase (also known as the immune tolerance phase) can persist for 10-30 years.

- This phase is characterised by the presence of hepatitis B e Antigen (HBeAg) and high hepatitis B virus DNA (HBV-DNA) levels with persistently normal alanine aminotransferase (ALT) levels. There are usually minimal histological changes in the liver. The rate of spontaneous HBeAg seroconversion is very low and 90% of children remain HBeAg-positive by the age of 10-15 years.

The second phase (also known as the immune clearance phase) usually occurs during late adolescence or young adulthood.

- This phase is characterised by elevated ALT levels, lower hepatitis B virus DNA levels and increased histological activity. This phase is often marked by sporadic ALT elevations thought to be associated with attempts by the immune system to eliminate hepatitis B virus infected hepatocytes. Spontaneous HBeAg seroconversion occurs at an annual rate of 10-20%. The level of ALT is an important predictor of seroconversion, with 60-70% of patients having spontaneous seroconversion if ALT levels are more than 5 times normal.
- The mean age of spontaneous HBeAg seroconversion is about 31-35 years with most (> 90%) patients seroconverting by the age of 40 years. Persistence of HBeAg and high hepatitis B virus DNA levels beyond this age often implies a poor prognosis with worsening histology and higher incidence of hepatocellular carcinoma.

The third phase (also known as the non-replicative phase) occurs when patients are usually asymptomatic and disease progression to cirrhosis is low if it has not occurred yet.

- This phase is characterised by low levels of hepatitis B virus DNA, absence of HBeAg (i.e., HBeAg negative) and presence of hepatitis B e antibodies (i.e., anti-HBe positive) with absence of hepatic inflammation histologically.

SCREENING & DIAGNOSIS

Screening:

As recommended by the Screening Test Review Committee (STRC), hepatitis B screening is recommended in the following individuals:

1. Asymptomatic Singapore residents with no known hepatitis B carrier status born before 1st September 1987 (when Hepatitis B vaccination was mandatory for all newborns) and who did not undergo the local catch-up immunisation programmes from 2001 to 2004 (conducted in secondary school, college and tertiary institution).
2. Pregnant women
3. Healthcare workers
4. Foreigners and immigrants from countries where HBV is endemic
5. **At risk groups** including but not limited to the below:
 - a) Chronic haemodialysis patients
 - b) Past or present injection drug users
 - c) Individuals who underwent invasive procedures in health-care facilities with inadequate infection control practices,
 - d) Individuals with known exposures to HBV (e.g., healthcare workers following needle stick injury involving HBV-positive blood, or recipients of blood or organs from a donor who tested HBV-positive)
 - e) Individuals whose past or present sex partners were / are HBV-infected or injection drug users or HIV patients
 - f) Other at risk groups at the discretion of the clinician

Under NUP guidelines, other patients who might be suitable for Hepatitis A or B screening are as below:

Who	When	Remarks
Patients who: • Have never had Hep A / B vaccination • Are uncertain they have had Hep A / B vaccination • Are uncertain of their Hepatitis A / B immune status	At most 6 months prior to going for Hep A / B vaccination	In some cases, patient might have screening results from more than 6 months ago and keen on vaccination. If patient is in the at-risk groups or clinically assessed to be at high risk for contracting Hepatitis B, it is recommended to screen for hepatitis B again before commencing vaccination ¹ to prevent delay or a missed diagnosis of chronic hepatitis B.

¹ Administration of Hepatitis B vaccination to a Hepatitis B carrier is not harmful and does not increase the risk for adverse events. The harm from vaccinating a high-risk patient who is already a hepatitis B carrier is that it will lead to a missed or delayed diagnosis of chronic hepatitis B.

NUP Hep A / B Screening and Vaccination Workflow

1. Before screening for hepatitis A / B, to check medical records that patient had not received hepatitis A / B vaccination before.
2. Arranged the appropriate screening tests as below.
 - Hepatitis A: Anti-HAV IgG*
 - Hepatitis B: HBs Ag, anti-HBs
3. Clinician will arrange for review when results are ready and offer the appropriate management. Refer to *section “Hepatitis B screening results and clinical interpretation”* for further guidance.
4. If the screening results was done more than 6 months ago, clinician to manage according to risk profile as listed below:
 - At risk groups: Repeat screening is recommended for at risk groups (see section on [Screening](#): at risk groups.)
 - Not at-risk groups: Repeat Hep B screening before vaccination is recommended but not mandatory. If the patient is not keen to repeat screening, counsel and proceed with vaccination if patient is agreeable.
5. To repeat Hep B screening with both HBs Ag and anti-HBs post-vaccination.

**Screening for Hepatitis A status (anti-HAV IgG) is not a mandatory requirement prior to vaccination but is available should patient request it.*

Hepatitis B Screening Results and Clinical Interpretation

Results HBs Ag	Anti-HBs	Vaccination status (Course of 3 doses)	Interpretation	Recommended Action
Negative (or non-reactive)	< 10 IU/L	No	Not immune to HBV	Administer hepatitis B vaccination.
Negative (or non-reactive)	< 10 IU/L	Completed recently within last few months	Not immune to HBV	Repeat hepatitis B course of 3 doses and recheck serology 6-8 weeks later. If no antibody response, consider referral to Infectious Diseases or Hepatology ²
Negative (or non-reactive)	< 10 IU/L	Completed many years ago	Antibody levels may have waned	Administer 1 dose and recheck Anti-HBs 6-8 weeks later. High titres > 100 convey immunity for life. If no antibody response, to complete course of 3 doses and recheck.
Negative (or non-reactive)	> 10 IU/L	Regardless	Immune to HBV	No vaccination required.
Positive (or reactive)	-	Regardless	HBV infection - either acute or chronic (carrier)	Look for signs and symptoms of acute hepatitis. Repeat HBs Ag in 6 months to check for Chronic Hep B infection. Screen for HIV and HCV.
Positive (or reactive)	> 10 IU/L	Regardless	HBV infection - either acute or chronic (carrier)	Look for signs and symptoms of acute hepatitis. Hep B carrier, possibly with the mutant variant ³ .

² These individuals might have Hep B virus infection with low viral and and an undetectable level of HBsAg. If total anti-Hep B Core is positive (which is not available as a test in NUP), they might be in the “window” phase of acute Hep B or they have chronic Hep B with low load viremia which requires specialty level management.

³ Also known as paradoxical co-existence of HBsAg and anti-HBs. Refer to hepatology for further management.

Hepatitis B Infection Diagnosis

1. Hepatitis B infection is diagnosed based on the presence of Hepatitis B Surface Antigen (HBs Ag).
2. Chronic HBV infection is defined as having two HBs Ag positive results taken at least 6 months apart.

Clinical Presentation

Acute hepatitis B is often asymptomatic or mild, especially in children under 5 years of age. In adults the onset of illness is usually abrupt and can last for weeks to months. Symptoms of acute hepatitis include:

- Jaundice
- Fever
- Dark coloured urine with pale stools
- Prolonged tiredness or malaise
- Poor appetite
- Abdominal pain
- Nausea and vomiting

Patients with acute hepatitis B⁴ need to be treated urgently as it can lead to acute liver failure and even death. Patients who recover fully from Hepatitis B infection usually develop immunity to Hepatitis B virus. However, 1 in 10 patients may develop Chronic Hepatitis B Infection (Hepatitis B carriers)

Patients with Chronic Hepatitis B Infection are usually asymptomatic unless they develop acute hepatitis or complications which can include liver cirrhosis, hepatocellular carcinoma (HCC) and liver failure. Therefore, screening for early diagnosis and regular follow-up with timely intervention is imperative to avoid the complications of chronic Hepatitis B.

⁴ Acute Hepatitis B is notifiable by laboratory to MOH. To include IgM anti-HBc if you suspect acute Hepatitis B. Chronic Hepatitis B does not require notification

Hepatitis B Viral Protein Tests and their Clinical Significance

Viral Protein Test	Clinical Significance
HBs Ag <i>Hepatitis B surface antigen</i>	Detected in high levels in serum during acute infection and persists for an average of 4 weeks after exposure to the virus. Persistence beyond 6 months indicates chronic HBV infection.
Anti-HBs <i>Hepatitis B surface antibody</i>	Indicates recovery and immunity from HBV infection. Also develops in a person successfully vaccinated against HBV.
IgM anti-HBc⁵ <i>IgM class antibody to core antigen</i>	Indicates recent infection with HBV (< 6 months) or acute flare of chronic Hepatitis B
HBe Ag <i>Hepatitis B envelope antigen</i>	Those positive for HBe Ag circulate HBV at very high titres in their blood. This indicates high infectivity. Persistence of HBe Ag beyond 40 years old is associated with poorer prognosis.
Anti-HBe <i>Antibody to HBe Ag</i>	Anti-HBe becomes detectable when HBe Ag is lost.

⁵ Consider adding IgM anti-HBc if there is clinical suspicion of acute Hepatitis B, eg new sexual partner, IVDU as acute hepatitis B is notifiable

MANAGEMENT

Management & Follow-Up of Chronic Hepatitis B Infection (for 16 years old and above)

Chronic Hepatitis B Infection – Persistent HBs Ag positive for at least 6 months (to repeat test if 1st screening) or **negative** IgM anti-HBc and a **positive** HBs Ag from the same blood sample

Initial Management

- **Family history** of HBV infection, HBV vaccinations, HCC and cirrhosis
- **Risk factors:** Smoking history, alcohol consumption, occupational history, medication history, history of jaundice and quality of life
- **Physical examination**
 - Stigmata of chronic liver disease - refer to GE SOC immediately if present
- **Investigations:**
 - Viral markers (HBe Ag, anti-HBe antibody)
 - LFT, AFP, U/S HBS, FBC

Physical examination and investigations normal

Continue 6 monthly follow-up

1. ALT, Bilirubin, Albumin, AFP, platelets 6 monthly
2. U/S HBS 6-12 monthly
3. HBe Ag - check once at age 40 or 35 years if high-risk factors present¹.
No need to repeat yearly subsequently.
4. Offer repeat Hep B serology (HBs Ag + Anti-HBs) every 2 years
 - This is to check for spontaneous seroconversion²
 - To consider continuing with regular surveillance even if there is HBsAg seroconversion as risk of HCC is still present.
5. Consider FIB-4 or APRI³ score for fibrosis assessment every 2 years

1. High risk factors: family history of HCC, regular alcohol consumption or immunocompromised state (prolong steroids usage, chemo / immunotherapy)
2. Presence of HBV antibodies
3. AST to platelet ratio index, not typically use in NUP. A score of ≤ 0.5 equal absence or minimal scarring

Any of the following:

1. Clinical signs of liver disease
 - Abdomen findings - hepatomegaly, splenomegaly, ascites
 - Extra abdomen findings - jaundice, spider naevi, leukonychia, asterix, pedal edema, palmar erythema*
2. Abnormal Laboratory results
 - a) ALT / AST persistently raised over 3 months
 - b) AFP raised OR rising trend OR HBe Ag +ve (> above age 40 or above age 35 if high risk factors present
 - c) Low albumin, raised bilirubin level, low platelet levels
3. U/S HBS suspicious for cirrhosis, HCC or abnormal lesions for which Computed Tomography (CT) is recommended

Refer gastroenterologist (GE SOC)

Management of Chronic Hepatitis B Infection

1. Monitor for acute flare and complications of Chronic Hepatitis B infection.

All patients should have a 6 monthly review to:

- a) Review for clinical symptoms and signs
- b) Liver Function Test – to look for acute liver inflammation or liver failure.
- c) Alpha-Fetoprotein (AFP) – Liver cancer marker
- d) Full Blood Count and Platelet – to evaluate liver function, to calculate APRI or FIB-4 score as a marker of liver fibrosis where indicated.
- e) Ultrasound Hepatobiliary system – to evaluate for abnormal changes such as fatty liver, fibrosis and growth.
- f) Frequency of review can be earlier according to clinician's assessment.

Specialist Referral Recommendation

Gastroenterology referral recommended in patients with:

- Persistently elevated ALT
- Abnormal Alpha-Fetoprotein level
- Other abnormal lab results: low albumin, raised bilirubin, low platelets
- Signs of liver cirrhosis¹, HCC or other abnormal lesions on U/S HBS
- Positive HBe Ag at 40 years old or 35 years old (for high-risk population) and beyond
- Clinical signs of chronic liver disease
- HIV or hepatitis C co-infection

Emergency department referral recommended in patients with:

- Clinical signs suggestive of acute liver injury or hepatic decompensation²

¹ Clinical signs include jaundice, hepatomegaly, ascites, pedal oedema and other stigmata of chronic liver disease (palmar erythema, spider naevi, telangiectasia, bleeding gums, purpura).

² Clinical signs include new onset of clinical jaundice, acute or overt gastrointestinal bleeding or ALT $\geq$ 1000 U/L.

2. Special situation:

- a) Offer repeat Hep B serology every 2 years to check for seroconversion (i.e., HBs Ag negative and anti-HBs positive)
- b) Sero-converted patients:
 - i. Discuss with patient to continue regular surveillance as risk of HCC is still present (although lower risk).
 - ii. Carry out fibrosis risk assessment with FIB-4 score every 2 years. Refer if Fib-4 score $>$ 1.3.
- c) Patients with acute flare where ALT is raised (see [NUP Metabolic Dysfunction-Associated Steatotic Liver Disease CPG](#))
- d) Fatty liver – to manage as per fatty liver workflow (see [NUP Metabolic Dysfunction-Associated Steatotic Liver Disease CPG](#))

Non-Pharmacological / Lifestyle Management

1. Encourage screening for metabolic risk factors and optimise cardiovascular risk factors as per population guidelines.
2. To stop smoking and stop intake of alcohol.
3. Advise to avoid use of Traditional Chinese Medication.
4. If patient is on prolonged immunosuppression medication (e.g., steroids, methotrexate), refer to a gastroenterologist for co-management as patient is at increased risk of chronic Hepatitis B flare.
5. Advise family members & especially the sexual partner for hepatitis B screening. Advise Hepatitis B vaccination for those who are not immune to Hepatitis B.
6. Reinforce on the usage of barrier contraception (e.g., condoms) during sexual intercourse with partner unless the partner is Hepatitis B immunised.
7. Reassure that Hepatitis B cannot be contracted by casual contact, sharing of utensils and sharing of common living space.
8. Where patient is high risk, assess for other sexually transmitted diseases such as Hepatitis C and HIV.

Self-Monitoring

Advise patients on signs and symptoms of acute flare and complications.

1. Jaundice
2. Abdomen pain
3. Weight loss or weight gain
4. Abdomen swelling
5. Loss of appetite
6. Pale stools with tea-coloured urine

RECOMMENDED CARE COMPONENTS FOR CHRONIC HEPATITIS B INFECTION

Recommended Care Component	Minimum Frequency*	Remarks
HBe Ag	At first visit	If positive at first visit, to recheck at age 40 years, or age 35 years if high-risk factors present ¹ ; if still positive, to consider specialist referral. Frequency of monitoring to be tailored based on clinical indication.
Anti-HBe antibody	At first visit	
Liver Function Test (LFT)	At first visit, and minimally ALT once every 6 monthly	Frequency of monitoring and specialist referral to be tailored based on previous ALT values and trends as well as HBe Ag status.
Alpha-fetoprotein (AFP)	At first visit and once every 6 monthly	AFP is a tumour marker used for HCC surveillance.
Full Blood Count (FBC)	Consider at first visit and once every 6 monthly	To monitor for thrombocytopenia associated with liver disease.
Ultrasound Hepatobiliary System	At first visit and annually	Frequency of imaging is based on HCC risk.
Hepatitis A Screening / Vaccination	Consider anti-HAV screening and vaccination	Unless contraindicated, hepatitis A vaccination should be given to prevent superimposed acute hepatitis A in patients with chronic hepatitis B virus infection.
Influenza Vaccination	Annually or per season	As recommended under the National Adult Immunisation Schedule (NAIS) and National Childhood Immunisation Schedule (NCIS).
Pneumococcal Vaccination (PCV13 or PPSV23)	As per guidelines depending on age and other medical conditions	As recommended under the National Adult Immunisation Schedule (NAIS) and National Childhood Immunisation Schedule (NCIS).
Sexually transmitted diseases and Hepatitis C screening	Screening in patients with high-risk behaviours ²	
Metabolic disease screening - blood pressure, lipid profile, weight and BMI, Diabetes	As per guidelines	Development of fatty liver and metabolic risk factors further increases risk of liver cirrhosis and HCC.

* More frequently if clinically indicated

¹ High-risk factors include family history of HCC, regular alcohol consumption or immunocompromised state such as history of HIV, long-term use of steroids, chemotherapy or immunotherapy.

² High-risk behaviours include men who have sex with men, unprotected sex with multiple sexual partners, injection drug users, tattoos, sharing of household articles contaminated with blood.

ABBREVIATIONS

AFP	Alpha-Fetoprotein
ALT	Alanine Transaminase
Anti-HBe	Antibody to HBe Ag
Anti-HBs	Hepatitis B surface antibody
FBC	Full Blood Count
HBe Ag	Hepatitis B envelope antigen
HBs Ag	Hepatitis B surface antigen
HBV	Hepatitis B Virus
HCC	Hepatocellular carcinoma
IgM anti-HBc	IgM class antibody to core antigen
LFT	Liver Function Test
U/S	Ultrasound

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